

Differentially Private Preference Data Synthesis for Large Language Model Alignment

Anonymous Authors¹

Abstract

Preference alignment constitutes an indispensable post-training stage through which large language models (LLMs) are induced to produce outputs congruent with human values. Its reliance on authentic human preference data, however, precipitates acute privacy concerns, inasmuch as such datasets routinely encode sensitive user prompts alongside the human judgments elicited from annotators. To reconcile alignment quality with privacy protection, we introduce **DPPrefSyn**, a novel algorithm that synthesizes differentially private (DP) preference data and thereby renders privacy-preserving preference alignment feasible. DPPrefSyn is a principled framework anchored simultaneously in the Bradley–Terry preference model and in the intrinsic geometric structure of pairwise human preference data: it first infers an underlying preference model from private data under formal differential privacy guarantees, and subsequently exploits this learned model, in conjunction with public prompts, to synthesize high-quality preference data. By capitalizing on the shared linear structure of per-cluster reward models, the method faithfully characterizes the heterogeneous human preferences embedded within private datasets, whereas its incorporation of DP Principal Component Analysis (DP-PCA) serves to sharpen learning accuracy. An extensive suite of experiments demonstrates that DPPrefSyn attains competitive alignment performance even under stringent DP guarantees, thereby underscoring the promise of synthetic preference data as a pragmatic surrogate for privacy-preserving preference alignment across a broad spectrum of applications. To the best of our knowledge, this work constitutes the first attempt to generate DP synthetic preference data for LLM alignment. Our code

will be released upon acceptance.

1. Introduction

Preference alignment techniques—most prominently RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024)—have become the predominant instruments for reconciling the behavior of LLMs with human expectations. Such algorithms are predicated on pairwise preference data, wherein human annotators contrast two candidate responses and designate the one that more adequately addresses a given prompt. The resulting preference data subsequently furnishes the supervisory signal for fine-tuning, whereby the model is encouraged to rank preferred responses above their less favored counterparts (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). It is precisely the demonstrated efficacy and reliability of preference alignment that have propelled its widespread adoption across applications as diverse as conversational assistants (Achiam et al., 2023), mathematical reasoning tools (Shao et al., 2024), and code generators (Shen et al., 2023).

The alignment of LLMs with human preference data nevertheless engenders substantial privacy concerns, precisely because such datasets so frequently incorporate authentic user prompts together with the feedback humans supply. The prompts themselves may divulge personal information pertaining to health, identity, or other sensitive matters, while the accompanying feedback may betray private beliefs, predilections, or behavioral regularities (Li et al., 2023; Yu et al., 2024). A body of prior work (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) has sought to attenuate these risks by invoking the rigorous safeguards afforded by differential privacy (DP) (Dwork et al., 2006b), and has reported encouraging results. Yet many of these approaches confer protection upon only a fraction of the data—privatizing, for instance, either the user prompts (Yu et al., 2024) or the preference labels (Chowdhury et al., 2024; Zhang et al., 2025), but seldom both in tandem. Such partial protection leaves residual privacy vulnerabilities, since an adversary might still infer sensitive attributes from whichever prompts or preference annotations remain unshielded. Compounding this limitation, these methods are

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

typically tailored to particular post-training algorithms such as RLHF (Wu et al., 2023a) and prove incompatible with more recent methods like DPO. They are, moreover, ordinarily constrained by the paucity of private preference data and frequently fall short of high-quality preference alignment, owing to the considerable cost and attendant scalability challenges of soliciting human preference annotations. These converging deficiencies motivate the central question that animates this work:

Is it possible to achieve superior preference alignment while simultaneously guaranteeing thoroughgoing privacy protection and preserving compatibility with emerging alignment methodologies?

The present work furnishes an affirmative response to this question. Our principal contributions may be summarized as follows:

- We introduce a new algorithm, DPPrefSyn (Figure 1), for generating DP synthetic preference data in service of LLM alignment. DPPrefSyn constitutes a principled framework grounded jointly in the Bradley–Terry preference model and in the intrinsic geometric structure of pairwise human preference data, and it is articulated through the following design elements. 1) *Capturing diverse human preferences via shared linear structure of rewards and clustering*. The Bradley–Terry preference model, coupled with a linear reward structure, permits us to partition preference samples according to the embedding differences between concatenated prompt–preferred and prompt–dispreferred pairs, and thereupon to approximate the preferences residing within each cluster by means of a shared reward function. 2) *Improving learning efficiency through DP-PCA*. So as to counteract the sample inefficiency and clustering instability that beset high-dimensional spaces, we deploy DP-PCA to project the embeddings into a lower-dimensional subspace while retaining, under DP, the salient preference signals. 3) *Saving privacy budgets by using public prompts*. We enlist public prompts to generate candidate responses and apply private reward models to adjudicate preference pairs, a stratagem that permits the privacy budget to be concentrated wholly on the modeling of private preferences and thereby enhances privacy efficiency.
- We furnish a rigorous privacy analysis of DPPrefSyn, thereby confirming that it adheres to (ϵ, δ) -DP. Within this analysis, we first apportion a privacy budget ϵ_0 to DP-PCA (Amin et al., 2019) for the purpose of dimensionality reduction, and ϵ_1 to DP-KMeans (Su et al., 2016) for clustering; the residual budget is then devoted to the training of per-cluster reward models via DP-SGD. By invoking the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the DP-SGD privacy expenditure tightly, we ensure that DPPrefSyn satisfies (ϵ, δ) -DP. By

virtue of the post-processing property of DP, the resulting synthetic dataset may be repurposed, without further privacy cost, across a multiplicity of preference alignment methods and distinct LLMs.

- We evaluate DPPrefSyn empirically on a battery of standard benchmarks, encompassing the question answering tasks drawn from OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022) as well as the TL;DR summarization task (Stiennon et al., 2020). Our experiments intimate that the synthetic preference data engendered by DPPrefSyn confers strong privacy guarantees while attaining competitive utility across a range of heterogeneous tasks. To illustrate, at $\epsilon = 2$ the fine-tuning of the Pythia-2.8B model with DPO on our synthetic data yields a GPT-4o win rate of 56.48% on Anthropic-HH, thereby markedly surpassing the 37.02% obtained by fine-tuning on real data. Such findings establish that our DP synthetic preference data can underwrite effective preference alignment of LLMs while upholding strong privacy guarantees. DPPrefSyn is, furthermore, amenable to a variety of preference optimization methods (e.g., DPO and RLHF) and retains its effectiveness under exacting privacy constraints, thus transcending the limitations of alternative privacy-preserving post-training approaches that remain wedded to specific methods.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) put forward a DP framework for aligning LLMs through reinforcement learning by adapting the PPO algorithm to the DP regime; their approach, however, remains confined to PPO, even as preference alignment continues to evolve through the advent of new methods. Chowdhury et al. (2024) investigate the estimation of rewards from preference-based feedback, appealing to the notion of label-DP (Ghazi et al., 2021) in order to safeguard the privacy of human annotators, while Zhang et al. (2025) advance AUP-RLHF, a user-level label-DP framework for RLHF. Effective though both methods are at protecting preference labels, neither addresses the sensitivity inherent in the prompts or responses themselves. These methods are, in addition, ordinarily constrained by the scarcity of private user preference data and may therefore fail to secure high-quality preference alignment, given the expense of procuring human preference annotations. Yu et al. (2024), by contrast, direct their attention to the protection of sensitive user instructions, generating DP synthetic instructions to supplant the real ones during data annotation and model fine-tuning.

Beyond preference alignment proper, a number of studies concentrate on the private fine-tuning of LLMs from general labeled data rather than preference comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024;

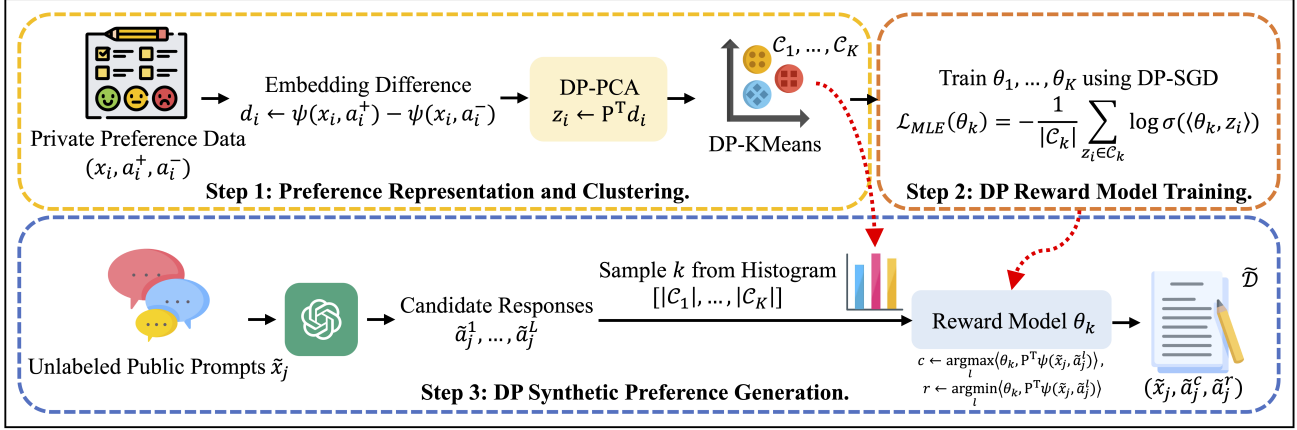


Figure 1. Overview of DPPrefSyn. DPPrefSyn synthesizes DP preference data by way of three successive steps: 1) it represents each preference sample as an embedding difference and clusters these differences via DP-PCA and DP-KMeans; 2) it trains a DP reward model on each cluster using DP-SGD; and 3) it generates synthetic preference samples from public prompts, guided throughout by the DP-protected distribution of reward functions.

Zhang et al., 2023; He et al., 2022), while a further recent strand of work is preoccupied with safeguarding the privacy of in-context prompts for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b).

Differentially Private Synthetic Text Generation. Our work situates itself within the broader ambit of DP synthetic text generation, a domain in which two representative lines of inquiry have crystallized. The first fine-tunes a language model on private data under DP and subsequently harnesses the fine-tuned model to generate synthetic text (Yue et al., 2023; Mattern et al., 2022; Mireshghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). Yu et al. (2024), for instance, privately fine-tune the LLaMA model on private instruction data and thereafter employ the resulting model to synthesize text for supervised instruction tuning.

Because DP fine-tuning can prove costly and, particularly for non-public language models, occasionally infeasible, a more recent line of work explores DP synthetic data generation through the prompting of LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). Such approaches typically generate synthetic samples via LLM API access and thereafter, in a privacy-preserving fashion, retain those outputs that most closely resemble the private data. Xie et al. (2024), for example, propose the Aug-PE algorithm, which iteratively refines LLM-generated samples on the basis of their embedding-space proximity to the private data. *Our work is consonant with this line of research and centers upon preference alignment for LLMs.*

Diversity in Human Preferences. A growing body of work underscores that human preferences are inherently heterogeneous (Denton et al., 2021; Aroyo et al., 2023a;b;

Chakraborty et al., 2024), with the salient contributing factors comprising sociodemographic disparities (e.g., race, gender, age), idiosyncratic beliefs and biases, unequal levels of domain expertise, and the constitutive ambiguity of natural language (Sandri et al., 2023; Vogels, 2021). Prompted by these observations, recent studies have set out to personalize preference alignment so as to reflect more faithfully the values of distinct user groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). Singh et al. (2025), to take one example, propose a meta-learning framework in which an LLM adapts to individual users from a modest number of preference examples. In the present work, we generate DP synthetic preference data that *captures the underlying diversity of human feedback*, and this fidelity to preference heterogeneity translates into improved alignment performance relative to approaches that disregard such diversity (as demonstrated in Section 5.3).

3. Problem Definition

Preference Alignment of LLMs. We concern ourselves with the preference alignment of an LLM on the basis of pairwise human feedback, formalized through a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each sample comprises a prompt x_i together with two responses a_i^+, a_i^- , of which a_i^+ is deemed preferable to a_i^- by human annotators.

In keeping with prior work on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we model human preferences by recourse to the Bradley–Terry model (Bradley & Terry,

1952):

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} = \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)),$$

where $a_i^+ \succ a_i^-$ signifies that a_i^+ is preferred to a_i^- , $r^*(x, a)$ denotes a latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This formulation constitutes the common substrate of both RLHF, which explicitly learns a reward model from preference data, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. Our objective is to shield the privacy of the preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against an adversary who endeavors to access or infer private information about individual prompts, responses, or preference labels from the outputs of the fine-tuned LLM. To this end, we propose to generate a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that conserves the critical preference information latent in $\mathcal{D}_{\text{priv}}$. By virtue of the post-processing property of DP, $\tilde{\mathcal{D}}$ may be reused across a variety of preference alignment methods and LLMs without incurring any additional privacy cost.

We formalize (ϵ, δ) -differential privacy by way of the following definition.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences within the private dataset are, as a rule, far from homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Different annotators may accord priority to disparate attributes of a response—factual accuracy, politeness, creativity, or clarity, among others—thereby giving rise to diverse and, at times, mutually conflicting signals within the data. Consequently, the faithful synthesis of $\tilde{\mathcal{D}}$ hinges critically on capturing the diverse human preferences embedded in $\mathcal{D}_{\text{priv}}$.

To render this tractable, we posit a linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Concretely, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

in which $\phi(x, a) \in \mathbb{R}^d$ constitutes the *shared* feature vector and $\theta \in \mathbb{R}^d$ constitutes a *user-dependent* parameter.

The linear structure strikes a felicitous balance between *expressiveness* and *tractability* in the modeling of diverse user preferences. Although individual users may harbor heterogeneous preferences, their reward functions tend to share common semantic features within a learned embedding space (i.e., $\phi(x, a)$) and to diverge principally in the weighting they assign to these features (i.e., θ). A linear parameterization of this kind apprehends the shared structure while accommodating variability across users or preference clusters, thereby enabling the effective modeling of preference heterogeneity under limited data and stringent privacy constraints. As our experimental results in Section B.6 attest, the linear reward structure captures preference signals effectively and, its simplicity notwithstanding, attains performance competitive with that of a fully fine-tuned GPT-2 reward model.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before expounding our design in full, we bring into relief several key components of the algorithm, each of which addresses one of the principal challenges attending the synthesis of differentially private preference data.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model entails that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

It follows that, for a tuple (x_i, a_i^+, a_i^-) , the relation $a_i^+ \succ a_i^-$ implies $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$; that is, the preference vector θ is strongly aligned with $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. For human annotators possessed of similar preferences, the feature difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ accumulated across their rated pairs may accordingly be expected to point in congruent directions within the embedding space. Guided by this intuition, we cluster the user preference samples on the basis of the feature difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$, in such a way that the preferences of users falling within a common cluster admit a faithful approximation by a shared reward function parameterized by a single preference vector θ . In Section B.9, we establish that the clusters so constructed are meaningful and interpretable with respect to user preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural candidate for the feature vector $\phi(x, a)$ is the embedding of the corresponding (prompt, response) pair, inasmuch as embeddings encode precisely those semantic similarities and differences that directly undergird human preference judgments. These embeddings are, however, ordinarily of high dimension (e.g., 1024 dimensions), the better to capture rich semantic and stylistic features. Learning in a space of such dimensionality typically demands a large sample

size—a demand that is inherently at odds with the limited number of samples available within any single cluster. To alleviate this tension, we apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022), which reduces the dimensionality of the embeddings and yields compact feature representations for clustering and reward model training. This step improves both sample efficiency and clustering stability while preserving the essential preference signals requisite for effective alignment.

Utilizing Public Prompts to Save DP Budget. The simultaneous synthesis of prompts, preferred responses, and less preferred responses can rapidly deplete the privacy budget and thereby erode utility. To circumvent this, we elect to draw upon public prompts, obviating any expenditure of privacy budget on the synthesis of DP prompts and allowing the entire budget to be marshaled toward modeling preferences over responses. For each public prompt, we generate multiple candidate responses with an LLM and, by applying a distribution of reward functions trained from private preference data under DP, assemble preference pairs. Although public prompts may be drawn from a distribution distinct from that of the private ones, our experiments reveal that, this prompt distribution shift notwithstanding, the synthetic preference data preserves the statistical preference distribution of the private preference dataset and remains effective for downstream preference alignment tasks.

4.2. DPPrefSyn Algorithm

We now set forth the DPPrefSyn algorithm (Algorithm 1), which synthesizes a DP preference dataset from the private dataset $\mathcal{D}_{\text{priv}}$. DPPrefSyn unfolds through three principal steps:

Step 1: Preference Representation and Clustering. We encode each prompt-response pair in $\mathcal{D}_{\text{priv}}$ by means of a public sentence embedding model ψ . Specifically, for each sample (x_i, a_i^+, a_i^-) we first concatenate the prompt with each response so as to form two texts, $[x_i; a_i^+]$ and $[x_i; a_i^-]$, whereupon we compute their embeddings via ψ and define their difference as $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

To effect dimensionality reduction privately, we apply DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, thereby obtaining a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ε_0 -DP (Line 5). Broadly speaking, DP-PCA (Amin et al., 2019) approximates the eigendecomposition of the data covariance matrix by estimating the collections of eigenvalues and eigenvectors separately in a DP manner. Each d_i is subsequently projected into a lower-dimensional space as $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). We then apply DP-KMeans clustering (Su et al., 2016) to $\{z_i\}_{i=1}^n$ in order to group samples exhibiting similar preference patterns under ε_1 -DP, thereby forming K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ (Line 7). In broad terms, DP-KMeans (Su et al., 2016) constitutes a DP rendition of the Lloyd algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, we learn the shared preference vector $\theta_k \in \mathbb{R}^p$ that encapsulates the cluster’s preference patterns by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i denotes the DP-PCA projected embedding difference obtained in Step 1. Each reward model is trained by means of the DP-SGD algorithm—a DP variant of SGD (Abadi et al., 2016)—employing a noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. We compute a histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ recording the number of samples in each cluster (Line 11), and normalize it so as to obtain a probability distribution $\mathbf{p}$ over the clusters, namely $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

Thereafter, for each public prompt $\tilde{x}_j$, we prompt an LLM at a relatively high temperature to generate L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). We then draw a cluster index $k \sim \mathbf{p}$ in accordance with the DP histogram (Line 15) and evaluate the responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ under the reward model parameterized by θ_k . To this end, we first compute the embeddings $\psi(\tilde{x}_j, \tilde{a}_j^l)$ for $l = 1, \dots, L$ and project them into the DP-PCA subspace via $\mathbf{P}$, whereupon the reward value assigned to $\tilde{a}_j^l$ is given by $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. We designate the responses attaining the highest and lowest reward values as the preferred and less preferred responses, respectively: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). Should the value gap prove too slight (e.g., < 0.5), we discard the sample so as to safeguard preference quality; otherwise, we append $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. Iterating this procedure over all public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Privacy Analysis of DPPrefSyn. We apportion a privacy budget ε_0 to DP-PCA (Line 5) and ε_1 to DP-KMeans (Line 7), reserving the remainder for the training of per-cluster reward models via DP-SGD with noise multiplier σ_2 (Line 9). For the training of the DP reward models we invoke the parallel composition property of DP: because the clusters are mutually disjoint, the modification of a single data point perturbs but one cluster, and the total privacy cost is thus bounded by the cost of training on the smallest cluster. We guarantee a known lower bound on the cluster size by discarding any cluster containing too few samples. We then employ the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) to compose the privacy costs incurred at each DP-SGD step, thereby ensuring satisfaction of $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. All remaining steps operate solely on public data or constitute post-processing, and hence occasion no additional privacy cost. A complete proof, together with the code for privacy accounting, is furnished in Sec-

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ 

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tion A.

Discussion on Prompt Distribution Shift. DPPrefSyn retains its effectiveness even when the prompt distributions of the private and public datasets diverge. The crucial observation is that the diversity of user preferences is captured through θ_k , the parameter of the linear reward model, rather than through $\phi(x, a)$, the feature vector associated with the prompt-response pair. To render this intuitive, consider that a human user may interact with an LLM through a wide variety of prompts even while their underlying preference remains invariant. It follows that, so long as we ensure the distribution of θ_k mirrors that obtaining in the private dataset, the resulting synthetic preferences will continue to reflect the private preference patterns. This is accomplished by drawing, for each public prompt, a cluster index k at random in accordance with the DP histogram (Line 15) and selecting the synthetic response pairs accordingly. Consequently, even where the prompt distributions of the private and public datasets differ, DPPrefSyn nonetheless preserves the preference patterns captured within the private data.

5. Experiments

Datasets. We conduct our evaluation on question answering and summarization tasks. For question answering, we draw upon the OpenAssistant dataset (Köpf et al., 2023), comprising assistant-style conversations, and the Anthropic-HH

dataset (Bai et al., 2022), which supplies human preference comparisons oriented toward helpfulness and harmlessness. For summarization, we recruit the TL;DR dataset (Stiennon et al., 2020), which furnishes human preference annotations over pairs of summaries. To emulate public data, we appropriate prompts from Alpaca (Taori et al., 2023) for the OpenAssistant QA task, from SafeRLHF (Ji et al., 2024) for the Anthropic-HH QA task, and from XSum (Narayan et al., 2018) for the TL;DR summarization task. Further details are available in Section B.2.

Setup for Algorithm 1. By default, we adopt BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ , fix the projected dimension at $p = 20$ in DP-PCA, and set the number of clusters to $K = 5$ in DP-KMeans. For DP-SGD, we employ a learning rate of 0.1, a batch size of 4, 4 training epochs, and a gradient clipping norm of 1.0. Candidate responses are generated with the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) at a temperature of 0.9, and we set the number of candidates per prompt to $L = 5$. We consider overall privacy budgets $\varepsilon = 0.5, 1, 2, 4, 8$ and fix $\delta = 1/|\mathcal{D}_{\text{priv}}|$. The influence of alternative generator models is examined in Table 3, and that of embedding models and hyperparameters in Section 5.3.

Evaluation. For downstream preference alignment, we fine-tune the Pythia-2.8B model (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), wherein the preferred response in the preference dataset serves as the training target, and thereafter apply the DPO algorithm (Rafailov et al., 2024) to fine-tune the SFT model further using preference pairs. Following Rafailov et al. (2024), we quantify the win rate of the model-generated responses against the preferred responses in the test set with the GPT-4o model. *Additional families of downstream models, along with alternative preference alignment methods, are investigated as ablation studies in Section 5.2.*

Baselines. We benchmark DPPrefSyn against the fine-tuning of the downstream model on real data via DP-SGD (Abadi et al., 2016), a baseline we denote DP-FT. It bears emphasis that DPPrefSyn is at once more reusable and more versatile than DP-FT, for it produces DP synthetic data that may serve to train diverse models under diverse alignment algorithms without any additional privacy cost, whereas DP-FT expends its privacy budget in the training of a single privatized model. We additionally incorporate a fully private baseline at $\varepsilon = 0$, in which the base model is evaluated without any preference fine-tuning whatsoever.

Particulars concerning the setups, hyperparameters, metrics, and baselines are provided in Section B.3.

5.1. Understanding the Performance of DPPrefSyn

In this section, we scrutinize the performance of DPPrefSyn by addressing two research questions bearing on its privacy-utility trade-off. Our principal results are marshaled in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? Across all tasks and for every privacy level $\epsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn consistently realizes a more favorable privacy-utility trade-off than DP-FT, and, while preserving privacy, it eclipses the utility of DP-unconstrained fine-tuning on real private data (DP-FT with $\epsilon = \infty$). On the OpenAssistant QA task at $\epsilon = 2$, for instance, our approach attains 9.10% after SFT and 11.04% after DPO, as against DP-FT’s 4.38% after SFT and 6.18% after DPO. Strikingly, and without any relaxation of its privacy guarantees, DPPrefSyn secures higher utility than the non-private baseline of fine-tuning on real private data (DP-FT with $\epsilon = \infty$), which reaches only 4.75% after SFT and 8.20% after DPO. An analogous pattern emerges on the Anthropic-HH dataset at $\epsilon = 2$, where DPPrefSyn achieves a win rate of 55.21% after SFT and 56.48% after DPO—against the 31.53% and 37.02% registered by DP-FT—and likewise surpasses direct fine-tuning on real data (DP-FT with $\epsilon = \infty$), which attains 31.98% after SFT and 38.72% after DPO.

These results bring into relief two **key advantages** of our DP synthetic data. **First**, DPPrefSyn engenders a pronounced performance improvement after SFT alone, in comparison with the direct application of SFT to private data. On the TL;DR summarization task at $\epsilon = 2$, for example, DPPrefSyn attains a win rate of 62.49% after SFT—far in excess of the 22.57% yielded by SFT on private data. This gain is attributable chiefly to the fact that DPPrefSyn generates candidate responses with a high-performance LLM, so that the resulting high-quality training data furnishes more efficacious supervision during SFT. This advantage persists even when public data is scarce: on the OpenAssistant QA task at $\epsilon = 4$ with a mere 5K public prompts, DPPrefSyn attains a win rate of 6.80% after SFT, as against 4.46% for SFT applied directly to the private data (Table 4), which comprises roughly 14K samples—a contrast that throws into relief DPPrefSyn’s capacity to contend with data scarcity while preserving utility. **Second**, applying DPO with our synthetic data atop the SFT model precipitates further improvements, an indication that our synthetic preference pairs encode meaningful distinctions aligned with human preferences. This is made possible by the private reward functions, which capture diverse user preferences while preserving privacy. Relative to the direct use of private data, the increment afforded by DPO is more modest in our setting, presumably because the SFT model trained on our synthetic data already performs strongly and thus leaves comparatively little room

for further optimization. Even so, when the fine-tuning pipeline is considered in its entirety, the performance gain accruing from our DP synthetic dataset remains substantially greater than that obtained from real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ϵ ? **First**, Table 1 reveals that DPPrefSyn generally outperforms DP-FT as ϵ ascends from 0.5, 1, 2, 4, 8 to ∞ , testifying to its favorable scalability with respect to the privacy budget ϵ . **Second**, DPPrefSyn can prove more resilient to DP noise than DP-FT in certain regimes. On OpenAssistant with Llama-3-8B as the downstream model, for instance, the DPO performance of DP-FT deteriorates from 59.13% to 45.79% as ϵ diminishes from ∞ to 4, whereas DPPrefSyn recedes only marginally, from 61.52% to 60.25% (see Table 3 in Section 5.2). A plausible explanation is that DP-FT perturbs the model parameters directly through DP-SGD, whereas DPPrefSyn injects noise during data synthesis, a mechanism that better safeguards utility.

5.2. Understanding the Properties of DPPrefSyn

In this section, we interrogate the properties of DPPrefSyn, encompassing its generalizability across disparate preference alignment methods and LLMs, its behavior under varying scales of public data, and its robustness to empirical privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn accommodates a variety of preference alignment methods and remains effective under stringent DP constraints. In Table 2, we evaluate DPPrefSyn in conjunction with SFT, DPO, and PPO. For PPO, we train K reward models—one per cluster—so as to capture the diverse preference patterns, and during PPO training we take the average output of the K reward models as the reward and optimize the policy accordingly. We find that, for SFT, DPO, and PPO alike, DPPrefSyn consistently secures superior utility relative to fine-tuning on private data, all the while preserving strong privacy guarantees.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn proves effective across a broad spectrum of downstream LLMs. In Table 3, we conduct experiments over a range of downstream models—Qwen-3-4B, Gemma-2-2B, and Llama-3-8B—employing Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generator models. We find that, after DPO, DPPrefSyn can outstrip fine-tuning on real data while safeguarding privacy across distinct model families. Using Qwen3-4B-Instruct-2507 as the generator, for example, DPPrefSyn achieves an 86.52% win rate on Qwen-3-4B at $\epsilon = 4$, thereby decisively surpassing non-private fine-tuning on real data (59.41%). We further observe that DPPrefSyn’s performance rises in tandem with the strength of the generator model, Qwen3-4B-Instruct-2507 outperforming

Table 1. GPT-4o win rates on the test sets for the downstream Pythia-2.8B model fine-tuned by SFT and DPO, where $\varepsilon = 0$ denotes the base (fully private, non-finetuned) LLM. *Relative to DP-FT, DPPrefSyn attains higher win rates across the privacy levels $\varepsilon = 0.5, 1, 2, 4, 8$, surpassing even the utility of non-private fine-tuning (DP-FT with $\varepsilon = \infty$).* Reported values are the mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. On OpenAssistant, DPPrefSyn attains higher win rates than fine-tuning on private data across SFT, DPO, and PPO alike, all while furnishing strong privacy guarantees. Reported values are the mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

Table 3. On OpenAssistant, DPPrefSyn attains higher win rates than fine-tuning on private data across a range of downstream LLMs after DPO, while furnishing strong privacy guarantees; it further stands to benefit from more powerful generator LLMs.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

Table 4. Across varying public data sizes m , DPPrefSyn attains higher win rates than fine-tuning on private data. Reported values are the mean and standard deviation over 5 random seeds.

Data Type (Size)	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

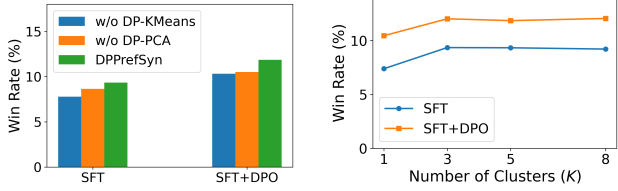
Llama-3-8B-Instruct.

RQ5: Can DPPrefSyn be effective with varying scales of public data? We examine the effect of the public data size m in Table 4. The private dataset (OpenAssistant) comprises 14K preference annotations, whereas the full public dataset (Alpaca) comprises 52K prompts. We randomly sample subsets of the public prompts at sizes $m = 5K, 10K, 52K$ and thereupon generate synthetic preference data with DPPrefSyn. We find that even with a mere 5K public prompts DPPrefSyn attains a win rate of 6.80% after SFT and 9.44% after DPO under $\varepsilon = 4$, thereby outperforming the DP-FT baseline, which attains only 4.46% and 6.12%, respectively. This demonstrates that **DPPrefSyn remains effective even where the available public data is limited**, a robustness owing to the high-quality responses generated by the LLMs and to the private reward functions that faithfully capture user preferences. Moreover, **as the public data size grows, the margin of improvement over the baseline becomes ever more pronounced**, an indication that DPPrefSyn can profitably harness the abundance of public data for preference alignment, thereby further attenuating the data scarcity that afflicts private preference data.

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? In Table 5, we mount membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the resulting AUC scores. We adhere to the prior work of Feng et al. (2025), in which the authors instantiate a MIA on pref-

Table 5. On OpenAssistant, DPPrefSyn exhibits lower AUC scores against MIAs (Feng et al., 2025), approaching the level of random guessing. Reported values are the mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64_{1.29}	49.98_{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51_{1.38}	50.43_{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45_{0.87}	50.67_{0.96}



(a) Ablating DP-KMeans or DP-PCA. (b) Varying the number of clusters.

erence data for LLM alignment; the adversary’s objective is to ascertain whether a given private sample (x, a^+, a^-) figured in preference alignment. We find that DP-FT fails to attenuate MIAs to a satisfactory degree, whereas **DPPrefSyn consistently exhibits lower AUC scores that verge on random guessing**. This intimates that DPPrefSyn is the more robust to empirical privacy attacks, conceivably because the synthetic character of the data used for fine-tuning intrinsically curtails the risk of overfitting to real private data. The attendant details are deferred to Section B.4.

5.3. Validating the Design of DPPrefSyn

In this section, we validate the design of DPPrefSyn by subjecting its algorithmic components to systematic scrutiny.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? We adjudicate this question by ablating either DP-KMeans or DP-PCA and by varying the number of clusters K , in each case on OpenAssistant with $\epsilon = 4$. **(i) Removing either DP-KMeans or DP-PCA.** Figure 2a discloses that the excision of either DP-KMeans or DP-PCA precipitates a performance decline. In the absence of DP-KMeans, all samples are consigned to a single cluster; the removal of DP-PCA, for its part, yields ineffective clustering and sample-inefficient learning within a high-dimensional space. Employing multiple clusters with both components in place improves performance over the single-cluster configuration. **(ii) Varying the number of clusters K .** In Figure 2b, we evaluate DPPrefSyn with $K = 1, 3, 5, 8$ clusters. As K grows, performance first improves at $K = 3$ and thereafter remains relatively sta-

ble at $K = 5, 8$ —an indication that DPPrefSyn is robust to K and that moderate values (e.g., $K = 3, 5$) strike an advantageous balance between diversity and data efficiency. Taken together, these results attest to the benefit of capturing diverse preferences through multiple clusters.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? In Section B.5, we conduct ablation studies over the embedding model ψ , the DP-PCA dimension p , the number of candidates per prompt L , the filter threshold, and the privacy budget allocation. Across a variety of sentence encoders—bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019)—DPPrefSyn remains uniformly effective. We find that $p = 20$ affords the best overall performance after DPO, that $L = 5$ constitutes a sound default choice, and that a moderate filtering threshold of 0.5 modestly outperforms no filtering, whereas a high threshold of 1 proves overly aggressive and thereby impairs performance. Finally, DPPrefSyn is robust to reasonable perturbations of the budget allocation ϵ_0 and ϵ_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To preclude confounding effects, we conduct controlled ablations in which all methods share an identical prompt set or an identical response generator. Even under these constraints, DPPrefSyn continues to surpass DP-FT (Appendix B.7), thereby demonstrating that **its performance gains are not merely an artifact of higher-quality public prompts or stronger response generators**.

6. Conclusion

In this work, we have introduced DPPrefSyn, an algorithm for synthesizing DP preference data from a private dataset. By modeling preference diversity through clustering, enhancing efficiency by means of DP-PCA, and conserving privacy budgets via public prompts, DPPrefSyn renders the synthesis of preference data for LLM alignment at once effective and private. Our experiments establish that DPPrefSyn upholds formal privacy guarantees while simultaneously outperforming fine-tuning on real data.

7. Limitations

DPPrefSyn presupposes LLM API access for the synthesis of preference data, an arrangement that may entail additional API costs. Our study confines itself to English text-based preference data, and the extension of DPPrefSyn to multilingual settings and multimodal data remains an open avenue for future inquiry. Furthermore, on account of computational constraints, our experiments encompass only models

of up to 8B parameters.

Impact statement

This work enables the preference alignment of large language models without the exposure of personal information, thereby helping to reconcile the benefits of large-scale data with the imperative of protecting individual privacy, promoting ethical data usage, and fostering societal trust in an ever more data-driven world.

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A. Privacy Analysis

In this section, we set out the privacy analysis for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We begin by introducing several concepts and pertinent theorems drawn from the literature that our analysis requires.

Many DP algorithms, the Gaussian mechanism among them, afford an entire family of (ε, δ) -DP guarantees: for each fixed ε there exists a corresponding $\delta(\varepsilon)$ such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

A signal advantage of employing privacy curves for DP mechanisms is that they admit composition guarantees tighter than those furnished by advanced composition theorems (Dwork et al., 2014). Privacy curves lend themselves to numerical composition, which yields the tightest guarantees available. Gopi et al. (2021) supply privacy curves for the composition of several standard mechanisms, including the Gaussian mechanism and the subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max \left\{ \varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i} \left(\frac{\delta_{\text{error}}}{k} \right) \right\}.$$

Gopi et al. (2021) also derive the privacy loss of a subsampled mechanism from that of the original mechanism, a result that may in turn be enlisted to bound the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\begin{aligned} \text{CDF}_{X_p}(t) &= \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases} \\ \text{CDF}_{Y_p}(t) &= \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases} \end{aligned}$$

We are now in a position to undertake the privacy analysis of our algorithm.

Proof. Our algorithm comprises three components that access the private dataset: a DP-PCA subroutine satisfying ε_0 -DP (Amin et al., 2019) in Line 5, a DP-KMeans subroutine satisfying ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In our experiments, we first apportion ε_0 to DP-PCA and ε_1 to DP-KMeans; the residual budget is thereafter assigned to DP-SGD and composed by means of the Privacy Random Variable accountant (Gopi et al., 2021).

In Line 7, we partition the low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint groups $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP, and in Line 9 a separate linear model is trained on each cluster via DP-SGD. By the parallel composition property of DP, the overall privacy cost of this step is governed solely by the smallest cluster employed for training. So as to guarantee a known lower bound on

the sample size within each model, we discard any cluster containing fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples. For DP-SGD, we fix the batch size at 4, the number of epochs at 4, and the gradient clipping norm at 1.0. The per-iteration privacy loss of DP-SGD conforms to the privacy curve of the subsampled Gaussian mechanism (Theorem A.4), and we thereupon invoke Theorem A.3 to compose the individual iterations of DP-SGD (Gopi et al., 2021), thereby ensuring satisfaction of $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. The DP parameters ϵ_0 , ϵ_1 , and σ_2 employed in our experiments are presented in Table 7.

The code underlying our privacy accounting is provided below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted on NVIDIA H100 GPUs, each endowed with 80 GB of memory. We execute DPPrefSyn on a single NVIDIA H100 GPU, a process that requires several hours to complete, the precise duration being contingent upon the dataset size.

B.2. Datasets

In this section, we delineate the datasets employed in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated and human-annotated assistant-style conversation corpus, encompassing 161,443 messages across 35 languages, together with 461,292 quality ratings and upward of 10,000 fully annotated conversation trees. In our experiments, we restrict ourselves to the English subset, which comprises 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) comprises human preference annotations used to train reward models for RLHF. For helpfulness, the data are partitioned into training and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered via rejection sampling with a preference model, and data amassed over the course of an online training process. For harmlessness, the data are gathered solely from base models but adhere to the same format. The dataset encompasses 161K training and 8.55K test comparisons, from the test set of which we sample 1,000 examples for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) comprises human preference annotations over pairs of summaries, wherein annotators are asked to select the superior of two summaries. It was originally devised to train a reward model for aligning summarization with human preferences. The summaries derive from the TL;DR dataset, while supplementary validation data are drawn from CNN and Daily Mail articles. The dataset encompasses 92.9K training and 86.1K validation comparisons, from the validation split of which we sample 979 examples as our test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) comprises 52K instruction-following examples generated through self-instruct with OpenAI’s text-davinci-003. Each example consists of an instruction paired with a corresponding response, the whole being designed to enhance instruction-following in language models. In this work, we appropriate the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset encompassing both performance and safety preferences. It imposes constraints along more than ten dimensions—among them insults, immorality, crime, emotional harm, and privacy—and is designed for fine-grained value alignment in RLHF. In this work, we appropriate the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for the evaluation of abstractive single-document summarization systems. The task consists in generating a one-sentence summary that answers the question, “What is the article about?” The dataset comprises 226,711 BBC news articles spanning 2010 to 2017, each paired with a concise summary, and traverses a broad range of domains, including politics, sports, business, science, and health. It encompasses 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we appropriate the documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

In this section, we delineate the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. Within DPPrefSyn, for each preference sample (x_i, a_i^+, a_i^-) we concatenate the prompt with each response so as to form two texts, $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The prompt-response concatenation format adopted in our experiments is exhibited below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we adopt BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . For DP-PCA we employ the implementation of Shoemate et al. (2021), which realizes the algorithm proposed by Amin et al. (2019), and for DP-KMeans the implementation of Holohan et al. (2019), which realizes the algorithm proposed by Su et al. (2016). We fix the projected dimension at $p = 20$ in DP-PCA and the number of clusters at $K = 5$ in DP-KMeans. When training the reward models via DP-SGD, we discard any cluster containing fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples, so as to guarantee a known lower bound on the sample size per model. For DP-SGD, we employ the SGD optimizer with a learning rate of 0.1, and we fix the batch size at 4, the number of epochs at 4, and the gradient clipping norm at 1.0. The DP-SGD implementation relies upon the Opacus library (Yousefpour et al., 2021).

By default, we enlist the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) to generate candidate responses for each public prompt, fixing the number of candidates per prompt at $L = 5$ and the generation temperature at 0.9. The prompt format employed for generating multiple completions is exhibited below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. In our experiments, we fix $\delta = 1/|\mathcal{D}_{\text{priv}}|$. The values of $|\mathcal{D}_{\text{priv}}|$ are provided in Table 6, while the DP parameters ε_0 , ε_1 , and σ_2 are presented in Table 7.

Table 6. Sizes of the private datasets employed in our experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. The DP parameters ε_0 , ε_1 , and σ_2 employed in the DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first perform supervised fine-tuning (SFT), wherein the preferred response in the preference dataset serves as the training target, and subsequently apply the DPO algorithm (Rafailov et al., 2024) to fine-tune the SFT model further by means of preference pairs. All models are trained at bfloat16 precision. Following Rafailov et al. (2024), for SFT we employ a learning rate of 5×10^{-7} and train for 1 epoch, whereas for DPO we set $\beta = 0.1$ and a learning rate of 1×10^{-6} and train for 3 epochs. The batch size is fixed at 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We adopt the win rate as our utility metric, which quantifies how frequently the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model at a temperature of 0.0 and compare it against the preferred response in the test set, with GPT-4o serving as the evaluator; the win rate is then the percentage of cases in which GPT-4o adjudges the model’s response to be the better one.

Our recourse to the win rate is motivated by its status as the standard evaluation protocol in recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), in which a strong LLM is enlisted as an automated judge to approximate human preference judgments. Prior work has established that high-quality frontier models render reliable and reproducible preference judgments strongly correlated with human annotations, while affording superior scalability and consistency relative to human studies (Zheng et al., 2023). We elect GPT-4o as the judge inasmuch as it constituted the most advanced model available at the time of our experiments. The prompt formats employed to generate the win rates are furnished below, in keeping with the format introduced by Rafailov et al. (2024).

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD by means of Opacus (Yousefpour et al., 2021) in conjunction with FSDP2 and Ghost Clipping. To streamline the privacy accounting, we stipulate that each user contributes to either SFT or DPO by partitioning the private dataset evenly into two disjoint halves, and we track the privacy loss with the PRV accountant. We conduct a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$, settling upon 0.1 for our final experiments. Learning rates are searched over $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, whereupon 5×10^{-5} is selected for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is adopted for Llama-3-8B. The batch size is fixed at 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B, and we train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

While DP furnishes a theoretical guarantee against privacy leakage, the assessment of empirical privacy risks is no less important. In this section, we mount membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the resulting AUC scores in Table 5.

We adhere to the prior work of Feng et al. (2025), in which the authors instantiate a MIA on preference data for LLM alignment; the adversary’s objective is to ascertain whether a given private sample (x, a^+, a^-) figured in preference alignment. We randomly designate 1,000 training examples as member data and 1,000 held-out test examples as non-member data. We find that DP-FT fails to attenuate MIAs to a satisfactory degree, whereas DPPrefSyn consistently exhibits lower AUC scores that verge on random guessing. This intimates that DPPrefSyn is the more robust to empirical privacy attacks, conceivably because the synthetic character of the data used for fine-tuning intrinsically curtails the risk of overfitting to real private data.

B.5. Ablation Studies on Hyperparameters

In this section, we conduct ablation studies over the embedding model ψ , the DP-PCA dimension p , the number of candidates per prompt L , the filter threshold, and the privacy budget allocation. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Varying the embedding model ψ . We conduct experiments over an array of sentence encoders, namely BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Our results in Table 8 indicate that, although embedding quality does bear upon performance, DPPrefSyn remains effective irrespective of the particular embedding model employed.

Table 8. Performance of DPPrefSyn under a range of embedding models ψ on OpenAssistant at $\varepsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We experiment with $p = 10, 20, 30$, as reported in Table 9. We find that $p = 20$ affords the best overall performance after DPO, thereby recommending itself as a sound default choice.

Table 9. Performance of DPPrefSyn under varying DP-PCA dimension p on OpenAssistant at $\varepsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . We experiment with $L = 3, 5, 8$, as reported in Table 10. We find that performance improves as L is raised from 3 to 5, yet $L = 8$ confers no further gains—an indication that $L = 5$ constitutes a sound default choice.

Varying filter threshold. We experiment with thresholds of 0, 0.5, and 1, as reported in Table 11. A moderate threshold of 0.5 modestly outperforms no filtering, inasmuch as it serves to expunge noisy or uninformative preference pairs, whereas a high threshold of 1 filters overzealously, curtailing data coverage and thereby precipitating a performance decline.

Varying privacy budget allocation. We conduct experiments to evaluate alternative apportionments of the budget between DP-PCA (ε_0) and DP-KMeans (ε_1). In the original configuration, we allocate $\varepsilon_0 = \varepsilon_1 = 0.5$ under a total privacy budget of $\varepsilon = 4$. We additionally evaluate two alternative allocations on the OpenAssistant dataset: (i) an allocation favoring DP-PCA

Table 10. Performance of DPPrefSyn under varying number of candidates per prompt L on OpenAssistant at $\varepsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Table 11. Performance of DPPrefSyn under varying filter threshold on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$), and (ii) an allocation favoring DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). The results, reported in Table 12, demonstrate that DPPrefSyn is robust to reasonable perturbations of the budget allocation.

Table 12. Performance of DPPrefSyn under varying privacy budget allocation on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

In this section, we expound our design choice of employing linear reward models in DPPrefSyn (Line 9). This design follows the Bradley–Terry model equipped with a linear reward function—an assumption widely adopted throughout the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023) and formalized in Section 3.

To appraise the validity of this modeling choice, we undertake an empirical comparison between our linear model and a fully fine-tuned GPT-2 reward model trained on the Anthropic-HH dataset. For each test sample (prompt, preferred response, dispreferred response), we feed both responses—each concatenated with the prompt—into the trained reward model. The reward model assigns a scalar to each input, and we count a prediction as correct whenever the preferred response is accorded a higher value than the dispreferred one; accuracy is then computed as the ratio of correct predictions to the total number of triplets. We find that the linear model attains 66% accuracy, as against 62% for the fully fine-tuned GPT-2 model—a result establishing that the linear reward model captures preference signals effectively and, its simplicity notwithstanding, performs competitively.

B.7. Ablation Studies on Prompt and Response Generator Quality

In this section, we conduct experiments to evaluate the respective effects of prompt quality and response quality upon the performance of DPPrefSyn. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Prompt Quality. We conduct experiments to ascertain whether the gains of DPPrefSyn are attributable to the superior quality of public prompts. Specifically, we rerun DPPrefSyn with the OpenAssistant prompts, rather than Alpaca, as its public prompt set; the DP-FT baseline, meanwhile, continues to draw upon its original private OpenAssistant dataset, so that both methods operate over *the same set of prompts* from OpenAssistant. The results are reported in Table 13.

Our results show that DPPrefSyn continues to outperform DP-FT even when both methods operate over the same OpenAs-

Table 13. Prompt quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

sistant prompts, an indication that its gains do not derive solely from higher-quality public prompts.

Candidate Response Quality. To ascertain whether the gains of DPPrefSyn stem from the use of stronger response generators, we introduce an enhanced DP-FT baseline. Specifically, we first employ *the same LLaMA-7B-Chat generator* as in DPPrefSyn to produce candidate responses for each prompt, then rank these responses by means of a trained reward model so as to construct synthetic preference pairs, and finally apply DP-FT to this strengthened synthetic preference dataset. This arrangement ensures that the enhanced DP-FT baseline is furnished with responses of quality comparable to those used in DPPrefSyn. The results are provided in Table 14.

Table 14. Response quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Our results show that, although the enhanced DP-FT baseline outperforms the original DP-FT, DPPrefSyn nonetheless attains still higher performance—an indication that the gains of DPPrefSyn cannot be reduced to the mere use of a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We conduct a direct head-to-head comparison in which GPT-4o adjudicates DPPrefSyn’s outputs against those of DP-FT on OpenAssistant, and we report the win rate of DPPrefSyn over DP-FT in Table 15.

Across the several privacy budgets, DPPrefSyn is preferred to DP-FT in upward of 70% of cases, thereby corroborating the consistent advantage witnessed in our main results.

Table 15. DPPrefSyn Win Rate over DP-FT on OpenAssistant.

ε	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

In this section, we assess whether the clusters discovered by DPPrefSyn correspond to meaningful preference groups. Because the datasets employed in our experiments carry no information regarding specific human annotator groups, it is difficult to interpret systematically which preference style each cluster identified by DP-KMeans represents. To surmount this difficulty, we simulate two groups of human annotators possessed of divergent preferences using GPT-4, following Chakraborty et al. (2024). Specifically, for each prompt drawn from the Alpaca dataset (Taori et al., 2023), we prompt GPT-4 to generate two responses under distinct instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is thus paired with these two responses, together with a simulated preference label and annotator type

(“elementary school student” or “PhD student”). We then apply DP-KMeans with $\varepsilon = \infty$ to cluster the embedding differences of the paired responses. We find that the resulting clusters align closely with the simulated preference groups: one cluster is dominated by PhD student preferences (95.6%), the other by elementary school student preferences (93.1%). These results intimate that the clusters are meaningful and interpretable with respect to user preference styles.

B.10. Samples from DPPrefSyn

Table 16. Samples generated by DPPrefSyn at $\varepsilon = 4$. In the OpenAssistant example, the preferred response is the more helpful and the better aligned with user intent; in the Anthropic-HH example, the preferred response is the more supportive and informative; and in the TL;DR example, the preferred summary is the more concise and the more sharply focused upon the main point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I’m here to help you with that. The sentence you provided has a few errors. Here’s the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: “finnished” should be “finished”; “resturant” should be “restaurant”. I hope this helps...

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I’m so sorry to hear that... adoption. It’s a very difficult and emotional decision... adoption is a legal process... It’s a permanent decision... consider the impact on your children’s lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I’m so sorry to hear that you’re considering adoption. It’s a very personal and emotional decision, and I’m here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

In this section, we present samples generated by DPPrefSyn, with representative examples for each task provided in Table 16. For the OpenAssistant task, the preferred responses are, as a rule, the more helpful, the more informative, and the better attuned to user intent: in the spelling correction example, the preferred response explicates each error, whereas the less preferred response merely enumerates the number of mistakes. For the Anthropic-HH task, though both responses are safe, the preferred responses nonetheless evince higher quality; in the adoption prompt, the preferred response addresses both the legal and the emotional dimensions, urges the seeking of professional help, and foregrounds child support, whereas the less preferred one is shorter and more formal, wanting in empathy and practical guidance. For the TL;DR summarization task, the preferred summaries are the more concise and the more sharply focused upon the main point, whereas the less preferred ones incorporate less pertinent detail: in the hotel fire article, for instance, the preferred summary foregrounds the salient facts concerning the fire and its impact, while the less preferred version admits extraneous information such as guest evacuation and travel rearrangements. These examples make plain that our synthetic data captures human preference patterns effectively. Further samples from DPPrefSyn are provided below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4. Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.

2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.

3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.

4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.

5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.

6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.

7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.

8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.

9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.

10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent

2. Sleeping bag

3. Sleeping pad

4. Backpack

5. Flashlight or headlamp

6. Water bottle or hydration bladder

7. Portable stove or camping grill

8. Cooking pot and utensils

9. Toiletries (toothbrush, toothpaste, wipes, etc.)

10. First aid kit

Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\varepsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\varepsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983. The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been “no hostage crises”, adding that the man was “alone in the room”.

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

“His firearm was taken away,” Dr Kutlubay said, adding that the gun in the officer’s possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year’s Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer’s possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It’s the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.